

Sandipan Nath

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Nationality: Indian · **Work Permit:** Germany

ABOUT ME

Computer Scientist with research experience in the nonlinear dynamics of spiking neural networks, reservoir computing, and reinforcement learning. Master's candidate in Data Science at Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU). Incoming Doctoral Researcher at Technische Universität Graz, specializing in the intersection of efficient, brain-inspired machine learning and deep learning.

EDUCATION

Technische Universität Graz

Graz, Austria

Ph.D. Candidate in Computer Science (Incoming)

Expected Start: 09/2026

- Non-Teaching research assistant position focusing on integration of Spiking Neural Network (SNN) models into wireless communication systems.
- Research Focus: Neuromorphic Computing, Spiking Neural Networks, Efficient AI.

Friedrich-Alexander-Universität Erlangen-Nürnberg

Erlangen, Germany

Master of Science in Data Science

01/04/2024 – Current

- Master's Thesis: *Effect of time-delayed coupling heterogeneity on stochastic resonance in neural networks* (Supervised by Prof. Dr. Marius Yamakou).
- Current Grade: 1.3
- Scale: Best 1.0, Worst 5.0

University of Alberta

Edmonton, Canada

Bachelor of Science with Specialization in Computing Science

04/09/2018 – 22/11/2022

- Final Grade: 3.34
- Scale: Best 4.0, Worst 0.0

WORK EXPERIENCE

Technische Universität Graz

Graz, Austria

Doctoral Researcher (Incoming)

Expected Start: 09/2026

- Project staff member at the Institute of Machine Learning and Neural Computation.
- *40 hours/week*

Siemens Energy Global A.G.

Erlangen, Germany

Automation and Data Science Developer (Working Student)

07/01/2025 – 15/07/2026

- Designed and implemented a centralized data management system to streamline data flow between various HVDC system design tools.
- Developed an interactive dashboard UI to manage tool interoperability via the database, incorporating user feedback.
- *15 hours/week*

University of Alberta

Edmonton, Canada

University Teaching Assistant

04/01/2021 – 20/04/2021

- Supervised practical labs, marked assignments/tests, and provided coursework assistance.
- *8 hours/week*

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, C++, C, Bash/Linux

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, Brian, NEST, NESTML, Gymnasium, OpenCV, NLTK, Reinforcement Learning, Reservoir Computing, CNNs, Neural Networks

High-Performance Computing: HPC Clusters, SLURM, Parallel Computing (Multiprocessing/Job Scheduling)

Data & Databases: Pandas, NumPy, MySQL, SQLite, MongoDB, FireBase

Tools & Platforms: Git, Jupyter Notebook, Azure, Matplotlib, Agile SCRUM

SELECTED PROJECTS

- Reinforcement Learning for Control Theory** | *Fraunhofer IIS* Completed
- **Key Skills:** Python, Reinforcement Learning, LSTM, Mixture Density Networks (MDN), Control Theory
 - Building a world model using LSTM + MDN to capture and predict complex system dynamics.
 - Training a controller to balance a ball on an arc (on a cart) based on the learned dynamics.
- Forecastability in Chaotic Neural Dynamics** | *FAU Erlangen-Nürnberg* Completed
- **Key Skills:** Reservoir Computing, Echo State Networks (ESN), Nonlinear Dynamics
 - Built an ESN to model Hindmarsh–Rose neurons in spiking, periodic bursting, and chaotic states.
 - Explored the network’s ability to forecast nonlinear and chaotic time series effectively.
 - Link: github.com/ND-SoSe24-Project/ND-Project
- Object Detection Models for Arctic Wildlife** | *University of Alberta* Completed
- **Key Skills:** Python, Computer Vision, PyTorch
 - Implemented state-of-the-art computer vision models like YOLO for the localization of caribou in camera traps.
 - Evaluated performance in Arctic conditions, focusing on accuracy and computational efficiency.
 - Link: github.com/Robertboy18/Object-Detection-Arctic-Wildlife
- Modelling Drug Effects in Tumor Cell Dynamics** | *FAU Erlangen-Nürnberg* Completed
- **Key Skills:** ODE Modeling, Systems Biology, SciPy, Numerical Methods
 - Fitted ODE models to describe cell growth, drug availability, and cell killing mechanisms.
 - Simulated and predicted tumor growth responses to various drug regimens.
 - Link: github.com/snath47/oncosys-M1
- Shortest Path in a Train Network** | *FAU Erlangen-Nürnberg* Completed
- **Key Skills:** Dijkstra’s Algorithm, Graph Theory, Data Structures, Python
 - Implemented Dijkstra’s algorithm to determine optimal routes using variable cost functions.
 - Adapted the system to handle multiple constraints such as time, distance, and transfer frequency.
 - Link: github.com/snath47/PathfindingIndianRail

HONOURS AND AWARDS

Murray Thomas Gibson Memorial Scholarship in Mathematics (University of Alberta) – 2019
University of Alberta Undergraduate Scholarship – 2019
Department of Mathematical Statistical Sciences Scholarship (University of Alberta) – 2018

LANGUAGE SKILLS

English: C2 (Proficient/Native level)
Hindi: C2 (Proficient/Native level)
Bengali: Native
German: B1.1 (Intermediate)

CERTIFICATIONS

GRE General Test (ETS): Mar 2024
Verbal Reasoning: 161, Quantitative Reasoning: 162, Analytical Writing: 3.5
Scales: Verbal Reasoning: 130-170, Quantitative Reasoning: 130-170, Analytical Writing: 0-6

TOEFL iBT (ETS): Jan 2024
Total Score 109/120 – Reading: 28/30, Listening: 30/30, Speaking: 23/30, Writing: 28/30

REFEREES

Prof. Dr. Marius Yamakou
Interim Full Professor (W3)
Department of Data Science (DDS)
Friedrich-Alexander-Universität
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Dr.-Ing. Tobias Feigl
Postdoctoral Scientist
Precise Localization and Data Analytics
Fraunhofer IIS
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Prof. Dr. Julio Vera-González
Head of "Laboratory of Systems Tumor Immunology"
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Uniklinikum Erlangen
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